

Davide Staub

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Education

Ph.D. in Earth Science and Engineering, Imperial College London

Oct 2025 – 2029 (expected)

Scalable Scientific Machine Learning Lab (supervisor: Dr. Ben Moseley). Year 1: cut the spectrum-recovery error of exoplanet measurements by more than half compared to state-of-the-art pipelines, in the first blind ground-truth benchmark on real data from the James Webb Space Telescope (JWST); both the method, NOVA, which infers spectra directly from raw detector pixels, and the benchmark are going into manuscripts. PhD project: calibrated machine-learning inference from raw detector data, from single-spectrum inference to joint three-dimensional retrieval of transmission, eclipse, and phase-resolved observations.

M.Sc. in Computational Science and Engineering, ETH Zürich

Sep 2021 – Mar 2024

Grade 5.9/6, with distinction; specialisation in scientific machine learning. Awarded the Willi Studer Prize for the best Master's degree in the programme. Thesis: *Solving the Elastic Wave Equation with Physics-Informed Neural Networks: A Robust and Critical Assessment*.

Semester Thesis, University of Oxford

Apr 2023 – Jun 2023

Built a working prototype physics-informed neural network for the elastic wave equation at the Department of Earth Sciences (PyTorch, JAX); the prototype became the basis of my Master's thesis.

B.Sc. in Computational Science and Engineering, ETH Zürich

Sep 2017 – Aug 2020

Grade 5.14/6; specialisation in robotics. Thesis: *A C++ Simulation of Micron-Scale Robots*.

Employment and Teaching

Graduate Teaching Assistant, MSc Deep Learning, Imperial College London

Oct 2025 – present

Supported a block course and ran weekly sessions for around 40 students, answering questions across the deep-learning syllabus and marking coursework.

Lecturer (external), Space Data, M.Sc. in Space Systems, ETH Zürich

May 2025 – present

Sole lecturer for the final block of the annual course, taught twice (2025 and 2026) to around 40 M.Sc. students each time: four hours per week over a month, on convolutional neural networks and U-Nets for denoising imagery of the Moon's permanently shadowed regions. Corrected two homework rounds and personally graded each three-student team's final five-page report.

Research Assistant, D-EAPS, ETH Zürich

Aug 2024 – present

Helped build the new core course *Space Data* from the ground up with Prof. Thomas Zurbuchen and Dr. Simon Stähler: researched and wrote course material and recruited the external lecturers who co-lead the course.

Research Assistant, HomanLab, University of Zürich

Aug 2024 – Sep 2025

Machine-learning research using LLM-based encodings to study how language is processed in the brain in health and psychosis; mentored undergraduate students. Results appear in the papers listed below.

Software Engineer, later COO (part-time), Blueocean Semantic Web Solutions GmbH

Jan 2017 – present

Boutique data-intelligence firm. Designed, built, and operated large-scale web, TikTok, and Telegram crawling systems (Python, OpenMPI) with LLM-based content analysis, leading a two-person engineering team and covering the full pipeline from infrastructure to client delivery, for clients including a national central bank and several cantonal police forces.

Machine Learning Engineer, UREGO AG

Oct 2021 – Aug 2022

Built a data-processing pipeline integrating CNNs, LSTMs, and YOLO to deliver personalised activity recommendations from large-scale user location and image data.

Machine Learning Intern, 42matters AG

Nov 2020 – Apr 2021

Engineered an automated app-icon matching pipeline, processing millions of icons with CNN-based cross-platform matching.

Private Tutor (Mathematics, Physics, Python)

Jan 2016 – Aug 2020

Publications and Preprints

- D. Staub**, F. Rabe, A. Misra, Y. Pauli, R. Hüppi, N. Yang, N. Lang, L. Michels, V. Edkins, S. Frühholz, I. Sommer, W. Hinzen, P. Homan. Coherence in the brain unfolds across separable temporal regimes. *arXiv:2512.20481*, 2025.
- N. Yang, R. He, P. Homan, I. Sommer, **D. Staub**, W. Hinzen. Cross-lingual robustness of LLM-brain alignment and its computational roots. *arXiv:2605.21049*, 2026.
- D. Staub**, B. Moseley. Solving the Elastic Wave Equation with Physics-Informed Neural Networks: A Robust and Critical Assessment. *arXiv:2609.07983*, 2026.

Manuscripts in Preparation

- D. Staub** et al. NOVA: inference of exoplanet transmission spectra from JWST observations.
In preparation, 2026.

Talks and Posters

- Contributed talk, British Seismology Meeting 2024: physics-informed neural networks for the elastic wave equation; featured in the meeting report in *Astronomy & Geophysics*.
- Poster: *Coherence in the brain unfolds across separable temporal regimes*. Alpine Brain Imaging Meeting, Champéry, January 2026 (presented by co-authors).
- Poster: *Coherence in the brain unfolds across separable temporal regimes*. SIRS Annual Congress, Florence, March 2026 (presented by co-authors).

Awards

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| Janet Watson Scholarship, Imperial College London | 2025 |
| Competitive PhD scholarship of the Department of Earth Science and Engineering, awarded to fund the doctoral research. | |
| Willi Studer Prize, ETH Zürich | 2024 |
| Awarded for the best Master's degree in Computational Science and Engineering. | |

Academic service

- Reviewer**, *Geophysical Prospecting* (2026)

Technical Skills

- Machine learning:** PyTorch and JAX; CNNs, U-Nets, LSTMs, transformers; physics-informed neural networks, including custom architectures; LLM internals (embeddings, tokenization, hidden-state and logit analysis); LLM-based neural encoding models for fMRI (ridge regression, permutation testing).
- Scientific computing:** Python, C++, Fortran, Taichi, Git; MPI/OpenMP parallelisation and GPU computing on PBS and Slurm clusters; inverse problems; linear and nonlinear optimisation (gradient descent, Newton, IRLS, trust-region methods); signal processing; numerical methods for PDEs (finite elements, finite volumes, finite differences, and others).
- Scientific data and modelling:** JWST calibration pipeline and NIRISS/SOSS detector systematics; custom detector-level forward models with injection-recovery validation; high-resolution fMRI; programmatic archive access (MAST, astroquery); FITS and HDF5 at scale.
- Languages:** German, English, French, Italian.

Selected Projects

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| Fractured-object reassembly, ETH Zürich | 2023 |
| PyTorch pipeline for 3D feature-point learning; adapted the SuperGlue feature-matching network to 3D, improving keypoint and descriptor quality over existing methods. | |
| Physically based snow simulation, ETH Zürich | 2022 |
| Material point method implementation in Taichi with a custom octree data structure for efficiency. | |